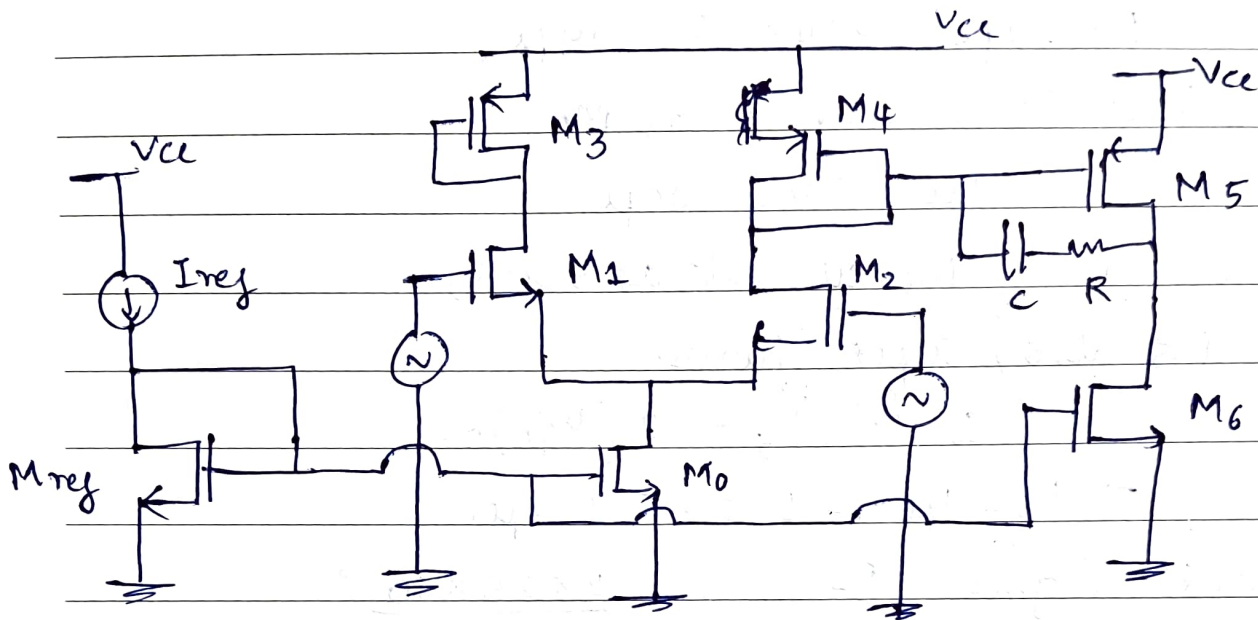


Analog 2 stage OTA

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EEE.

let $I_{ref} = 5\mu A$ and since tail current = $10 I_{ref}$
 $= 50\mu A$.



current to $M_1 = 25\mu A$

assume overdrive for $M_1, M_2 = 100mV$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$\Rightarrow 25\mu = \frac{1}{2} (230\mu) \left(\frac{W}{L}\right) (0.1)^2 \Rightarrow \left(\frac{W}{L}\right)_{M_1, M_2} = 21.74$$

choose $L_1 = 1\mu$ $W_1 = 21.74\mu \Rightarrow L_2 = 1\mu$ $W_2 = 21.74\mu$

M_{ref} : $L = 1\mu$, $W = 70\mu$

M_0 will have W/L 10 times the M_{ref}

$$\Rightarrow L_0 = 1\mu, W_0 = 100\mu$$

$$\text{for } M_3, I_D = \frac{1}{2} (\mu_p C_{ox}) \left(\frac{W}{L}\right) (V_{GS} - V_{thp})^2$$

$$\Rightarrow 2 \times 2.5 \times 10^{-6} = 100 \times 10^{-6} \left(\frac{W}{L}\right) (V_{GS} - V_{thp})^2$$

Date: / / $\Rightarrow$ But $V_{GS} - V_{thp}$ of pmos is:

$$1.8 - 1 - 0.39V$$

$$= 0.41V$$

$$50 \times 10^{-6} = 100 \times 10^{-6} \left(\frac{W}{L} \right) (0.4)^2$$

$$\frac{W}{L} \approx 2.974$$

choose $L_3 = 7\mu$ $W_3 = 2.974\mu$

$L_4 = 1\mu$ $W_4 = 2.974\mu$

M_6 has same current as M_0

$\Rightarrow L_6 = 1\mu$ $W_6 = 100\mu$

as we are using M_6 's current as $50\mu A$,
for M_5 also, $50\mu A$ flows.

so $(W/L)_5 = 2 (W/L)_3$

$\Rightarrow L_5 = 1\mu$, $W_5 = 5.948\mu$

$$\text{theoretical } g_m = \frac{2I_1}{V_{GS} - V_{th}} = \frac{2 \times 25 \times 10^{-6}}{10^{-1}}$$

$$= 5 \times 10^{-4}$$

Simulated $g_m \approx 4.2 \times 10^{-4}$

$$\begin{aligned} r_{o1} &= \frac{1}{E_{ids1}} = \frac{1}{1.00 \times 10^{-6}} = 10^6 \\ r_{o2} &= \frac{1}{2.99 \times 10^{-6}} = 3.3 \times 10^5 \end{aligned} \quad \left. \vphantom{\begin{aligned} r_{o1} &= \frac{1}{E_{ids1}} = \frac{1}{1.00 \times 10^{-6}} = 10^6 \\ r_{o2} &= \frac{1}{2.99 \times 10^{-6}} = 3.3 \times 10^5 \end{aligned}} \right\} \text{simulated.}$$

$$r_{o1} \parallel r_{o2} = 2.48 \times 10^5$$

$$\begin{aligned} \text{gain of stage 1} &= g_{m1} (r_{o1} \parallel r_{o2}) \\ &= 4.2 \times 10^{-4} \times 2.48 \times 10^5 \\ &= 104.16 \end{aligned}$$

$$20 \log(104.16) \approx 40 \text{ dB}$$

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gain of stage 2:

$$g_{m5} = \frac{2I_5}{V_{GS} - V_{th}} = \frac{2 \times 50 \times 10^{-6}}{1 - 0.418} = 1.71 \times 10^{-4}$$

~~Simulated $g_{m5} = \frac{100 \times 10^{-6}}{0.502}$~~

Simulated $g_{m5} = 1.63 \times 10^{-4}$

$$r_{o5} = \frac{1}{g_{d5}} = \frac{1}{2.39 \times 10^{-6}} = 4.18 \times 10^5$$

$$r_{o6} = \frac{1}{6.47 \times 10^{-6}} = 1.54 \times 10^5$$

} Simulated

$$r_{o5} \parallel r_{o6} = 1.12 \times 10^5$$

$$\begin{aligned} \text{gain 2} &\approx g_{m5} (r_{o5} \parallel r_{o6}) \\ &\approx (1.63 \times 10^{-4}) (1.12 \times 10^5) \\ &= 18.256 \end{aligned}$$

$$\begin{aligned} \text{net gain} &= (\text{gain 1}) \times (\text{gain 2}) \\ &= (104.16) (18.256) = 1901.54 \\ &= 65.58 \text{ dB} \end{aligned}$$

we took $C_2 = 5 \text{ pF}$, ~~by~~ due to miller effect,

$$\begin{aligned} (C_2)_{\text{input side}} &= (1 + A) C_2 \\ &= (1 + 18.256) (5 \text{ pF}) \\ &= 96.28 \text{ pF} \end{aligned}$$

$$(C_2)_{\text{output side}} = (1 \text{ pF}) + 5 \text{ pF} = 6 \text{ pF}$$

$$\begin{aligned} \# \text{ Pole 1: } \omega_{\text{Pole 1}} &= \frac{1}{C_{\text{eq}} (C_2)_{\text{input}}} = \frac{1}{2.48 \times 10^{-5} \times 96.28 \text{ pF}} \\ &= 2118.17 \end{aligned}$$

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Pole 2:-

$$\frac{1}{(R_{2eff})(C_2)_{output}} = \frac{1}{1.12 \times 10^5 \times 6 \times 10^{-12}}$$

$$= 0.14 \times 10^7$$

$$= 1400 \text{ kHz}$$

		Length (μm)	Width (μm)
M0	Mref	1	10
M1	M0	1	100
	M1	1	21.74
	M2	1	21.94
	M3	1	2.94
	M4	1	2.94
	M5	1	5.948
	M6	1	100

Mref : $V_{gs} - V_{th} = 0.456 - 0.408 = 0.048$

$V_{DS} = 0.456$ $V_{DS} > V_{gs} - V_{th} \Rightarrow$ Saturation

M0 : $V_{gs} - V_{th} = 0.456 - 0.409 = 0.047$

$V_{DS} = 0.316 \Rightarrow V_{DS} > V_{gs} - V_{th} \Rightarrow \text{SATN}$

M1 : $V_{gs} - V_{th} > V_{DS}$, $0.384 - 0.3 < 0.5 \Rightarrow \text{sat}$

M2 : $0.584 - 0.5 < 0.484 \rightarrow \text{sat}$

M3 : $V_{gs} - V_{tp} = 0.897 - 0.417 < 0.974 (V_{SD})$
 $\Rightarrow \text{sat}$

M4 : $V_{gs} - V_{tp} = 1 - 0.418 = 0.582$
 $V_{SD} = 0.904$ } sat

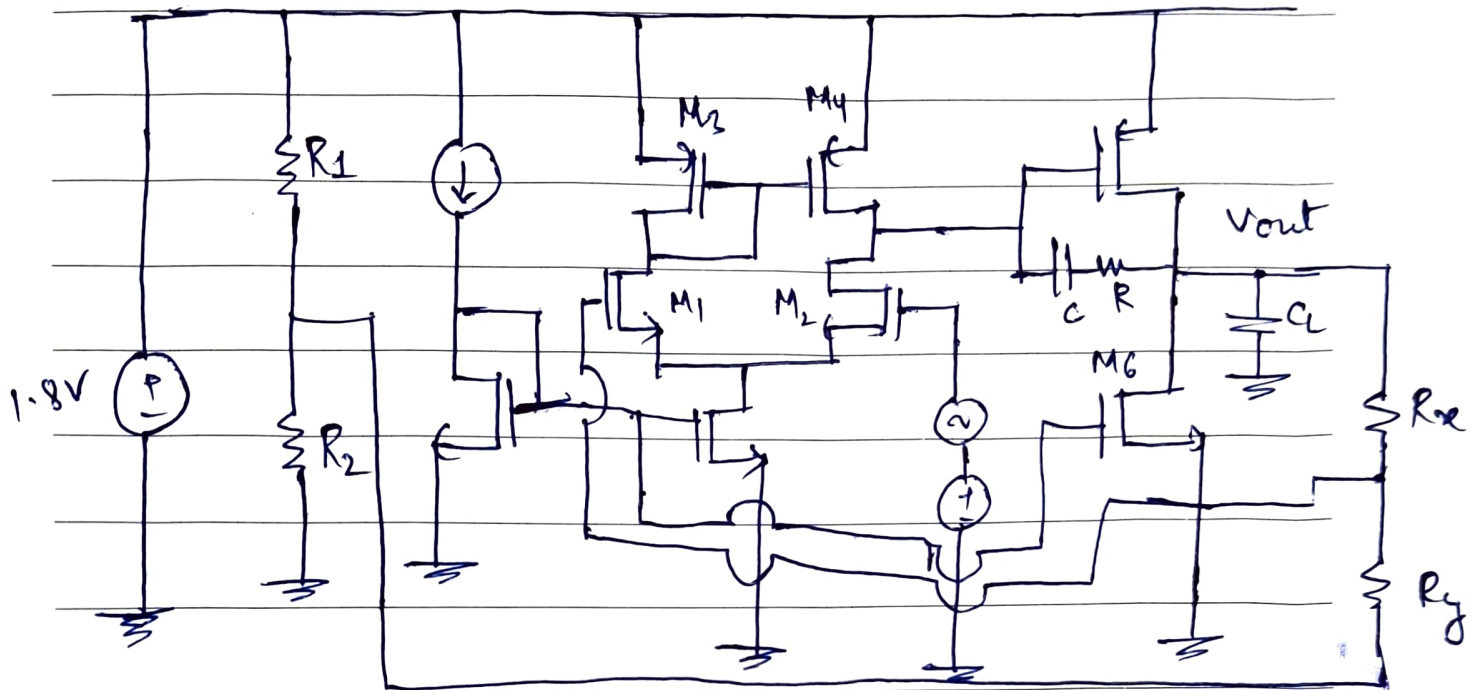
Date: / / M5 : $V_{gs} - V_{tp} = 1 - 0.418 = 0.582$
 $V_{SD} = 0.904$
 $\Rightarrow \text{satn.}$

M6: $V_{SG} - V_{th} = 0.456 - 0.405 = 0.05$

$V_{SD} = 0.899V \Rightarrow \text{satn.}$

$\Rightarrow$ all MOSFETs are in saturation $\checkmark$.

Final design:



I have used R_1, R_2 as voltage divider to get $0.9V$

$R_x = R_y = 2000k$

phase at $0dB = -103.5^\circ$

$\Rightarrow$ phase margin $= 180 - 103.5 = 76.5^\circ$

It's good enough since I also have to put constraints on gain bandwidth and phase margin and the system is stable,

$C_L = 13.55pF$ $C = 50.55pF$

~~Q~~

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